```
import pandas as pd
```

```
data_dict={
    'name':['arun','alice','ben','bob','john','luke','david','irin','ja
    'age':[21,23,23,24,30,34,32,38,34,23,33,22,21,20,36,27,33,38,29,25]
    'department':['IT','HR','HR','HR','HR','HR','IT','IT', 'admin', 'ad
    'salary':[2500,3000,2700,3300,3500,6000,4500,4300,5000,2500,3000,47
    'experience':[2,3,5,4,3,6,3,3,4,2,5,4,3,2,3,3,4,4,5,3],
    'performance score':[6.5,8.1,8,8,7,7.8,7.6,8.5,7.8,7.6,7.5,7,8,6,9.
}
df=pd.DataFrame(data_dict)
print(df)
```

```
       name  age department  salary  experience  performance score
0      arun   21         IT    2500           2                6.5
1     alice   23         HR    3000           3                8.1
2       ben   23         HR    2700           5                8.0
3       bob   24         HR    3300           4                8.0
4      john   30         HR    3500           3                7.0
5      luke   34         HR    6000           6                7.8
6     david   32         IT    4500           3                7.6
7      irin   38         IT    4300           3                8.5
8      jack   34      admin    5000           4                7.8
9       joe   23      admin    2500           2                7.6
10     zara   33         MD    3000           5                7.5
11    zeira   22         MD    4700           4                7.0
12      anu   21         MD    6700           3                8.0
13     sree   20         IT    3000           2                6.0
14      zam   36         IT    3300           3                9.5
15     neha   27         HR    4500           3                7.0
16     asha   33         IT    4800           4                6.0
17  Charlie   38         MD    3600           4                8.0
18     evan   29      admin    2800           5                8.1
19      tom   25         HR    3000           3                8.5
```

```
import numpy as np
```

```
salary_array=np.array(data_dict['salary'])
exp_array=np.array(data_dict['experience'])
print(salary_array)
print(exp_array)
```

```
[2500 3000 2700 3300 3500 6000 4500 4300 5000 2500 3000 4700 6700 3000
 3300 4500 4800 3600 2800 3000]
[2 3 5 4 3 6 3 3 4 2 5 4 3 2 3 3 4 4 5 3]
```

```
salary=np.array([2500,3000,2700,3300,3500,6000,4500,4300,5000,2500,3000,
mean=np.mean(salary)
median=np.median(salary)
```

```
std_deviation=salary.std()
print("mean:",mean)
print("median:",median)
print("std_deviation:",std_deviation)
```

```
mean: 3835.0
median: 3400.0
std_deviation: 1150.7714803556787
```

```
experience=np.array([2,3,5,4,3,6,3,3,4,2,5,4,3,2,3,3,4,4,5,3])
min_exp=np.min(experience)
max_exp=np.max(experience)
print("min_exp:",min_exp)
print("max_exp:",max_exp)
```

```
min_exp: 2
max_exp: 6
```

```
salary=np.array([2500,3000,2700,3300,3500,6000,4500,4300,5000,2500,3000,
min_sal=np.min(salary)
max_sal=np.max(salary)
normalise=(salary-min_sal)/(max_sal-min_sal)
print("normalise:",normalise)
```

```
normalise: [0.         0.11904762 0.04761905 0.19047619 0.23809524 0.8333
 0.47619048 0.42857143 0.5952381  0.         0.11904762 0.52380952
 1.         0.11904762 0.19047619 0.47619048 0.54761905 0.26190476
 0.07142857 0.11904762]
```

```
df=pd.DataFrame(data_dict)
print(df.head(5))
```

```
    name  age department  salary  experience  performance score
0   arun   21         IT    2500           2                6.5
1  alice   23         HR    3000           3                8.1
2    ben   23         HR    2700           5                8.0
3    bob   24         HR    3300           4                8.0
4   john   30         HR    3500           3                7.0
```

```
print(df.describe())
```

```
             age       salary  experience  performance score
count  20.000000    20.000000   20.000000          20.000000
mean   28.300000  3835.000000    3.550000           7.625000
std     6.087952  1180.666627    1.099043           0.854015
min    20.000000  2500.000000    2.000000           6.000000
25%    23.000000  3000.000000    3.000000           7.000000
50%    28.000000  3400.000000    3.000000           7.800000
75%    33.250000  4550.000000    4.000000           8.025000
max    38.000000  6700.000000    6.000000           9.500000
```

```
avg_sal=df.groupby('department')['salary'].mean()
print(avg_sal)
```

```
department
HR       3714.285714
IT       3733.333333
MD       4500.000000
admin    3433.333333
Name: salary, dtype: float64
```

```
ps=df.loc[df['performance score'].idxmax()]
print(ps)
```

```
name                  zam
age                    36
department             IT
salary               3300
experience              3
performance score     9.5
Name: 14, dtype: object
```

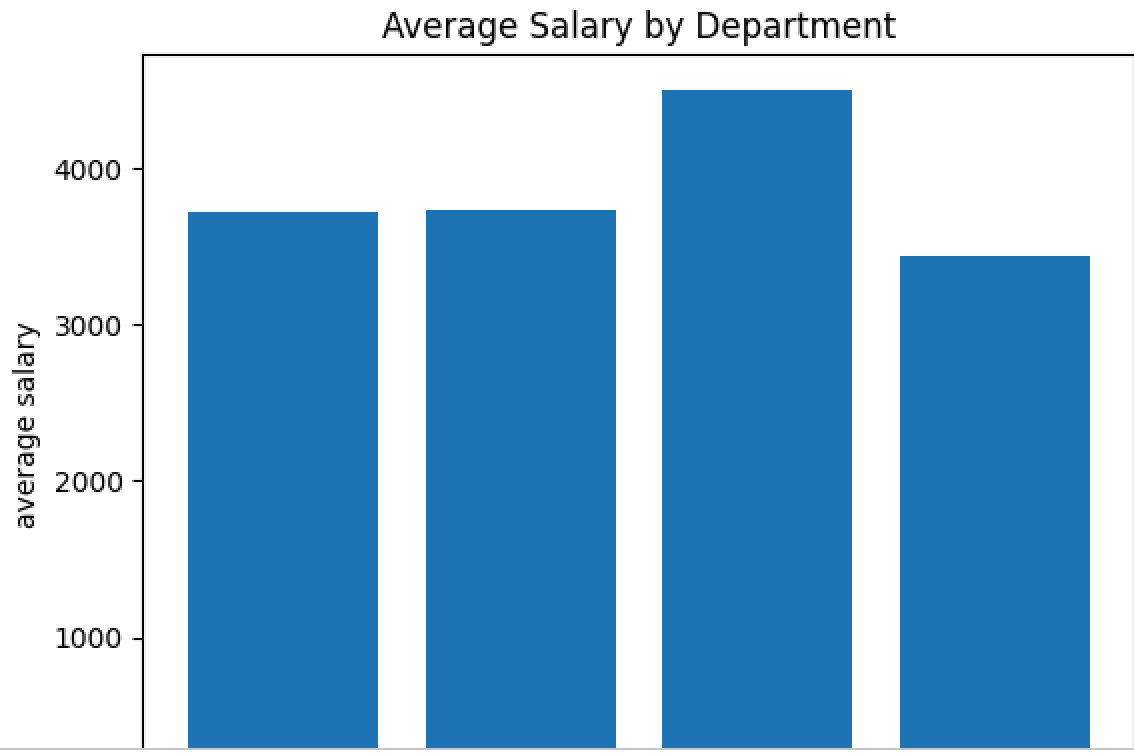
```
df['salary_after_bonus']=df['salary']*1.10
print(df['salary_after_bonus'])
```

```
0     2750.0
1     3300.0
2     2970.0
3     3630.0
4     3850.0
5     6600.0
6     4950.0
7     4730.0
8     5500.0
9     2750.0
10    3300.0
11    5170.0
12    7370.0
13    3300.0
14    3630.0
15    4950.0
16    5280.0
17    3960.0
18    3080.0
19    3300.0
Name: salary_after_bonus, dtype: float64
```

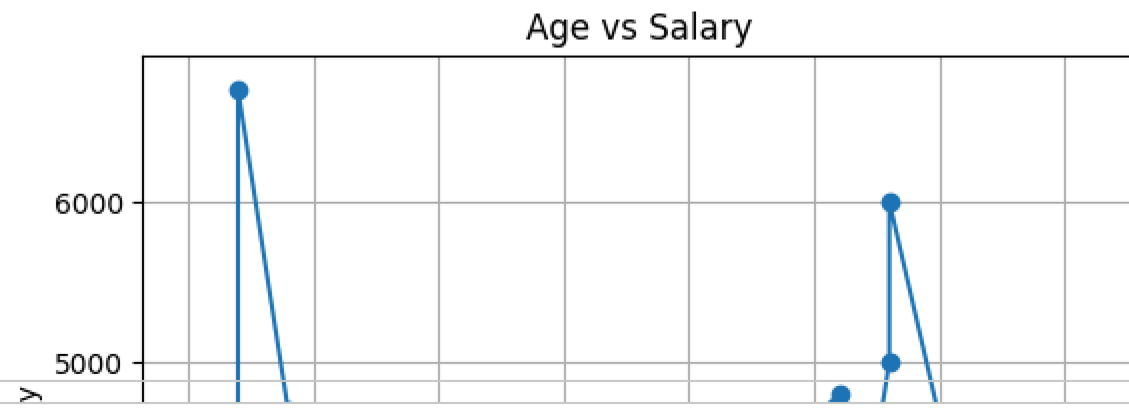
```
import matplotlib.pyplot as plt
```

```
avg_sal=df.groupby('department')['salary'].mean()
plt.bar(avg_sal.index,avg_sal.values)
plt.title("Average Salary by Department")
```

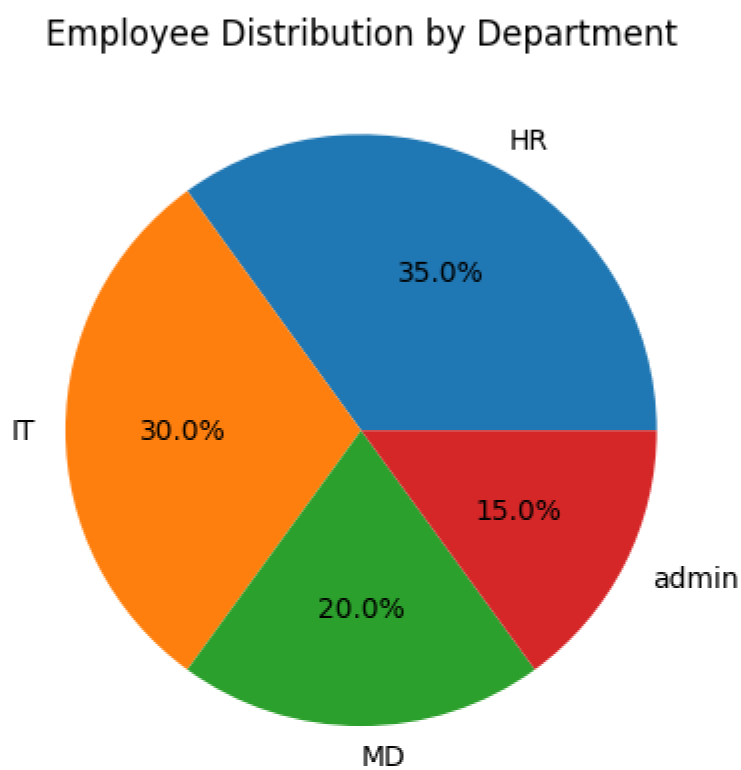
```
plt.xlabel("department")
plt.ylabel("average salary")
plt.figure(figsize=(6,4))
plt.show()
```

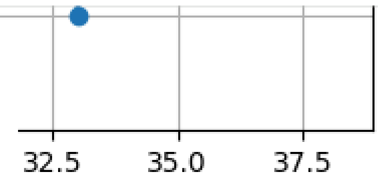



```
df_sorted = df.sort_values('age')

plt.plot(df_sorted['age'], df_sorted['salary'], marke
plt.title("Age vs Salary")
plt.xlabel("Age")
plt.ylabel("Salary")
plt.grid(True)
plt.figure(figsize=(6,4))
plt.show()
```



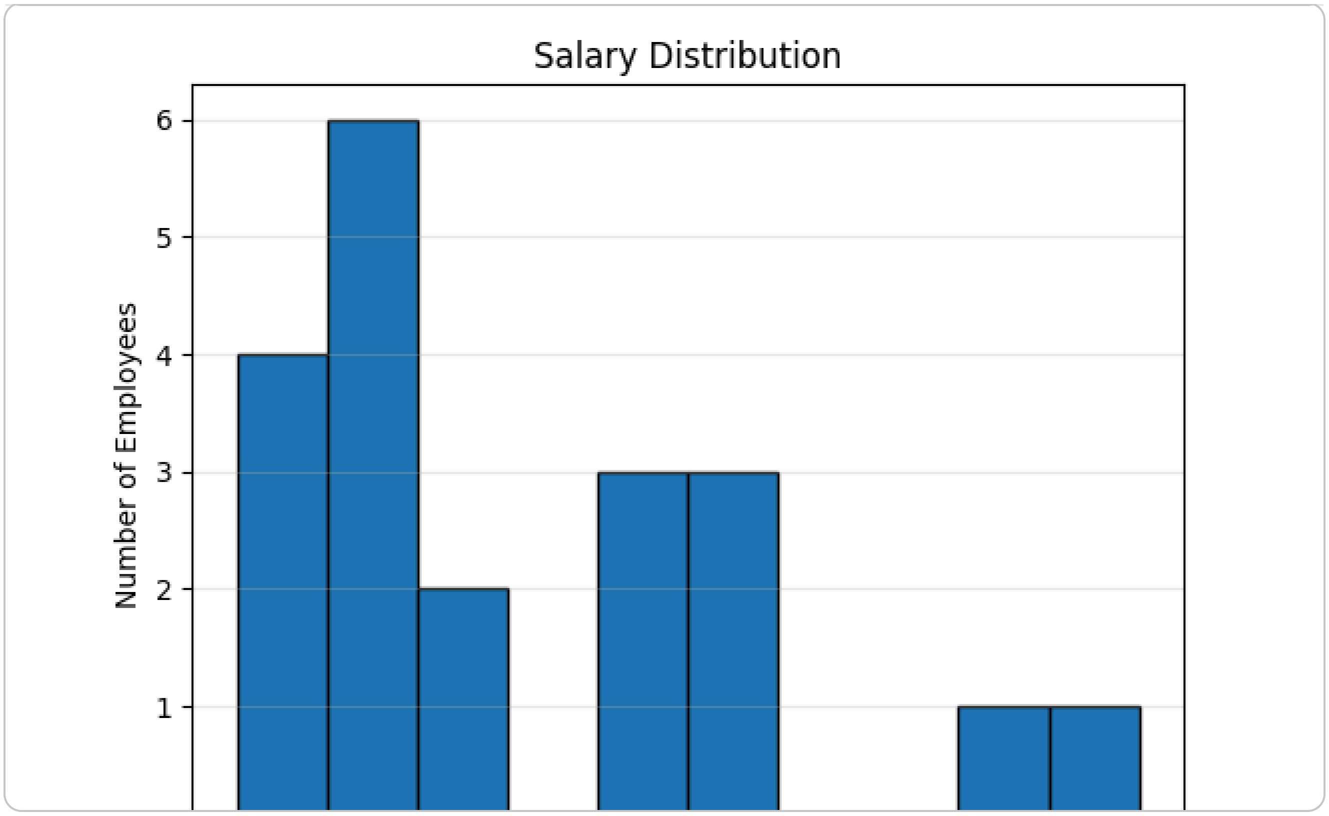

```
dept_counts = df['department'].value_counts()

plt.pie(dept_counts, labels=dept_counts.index, autop
plt.title("Employee Distribution by Department")
plt.figure(figsize=(11.69,8.27))
plt.show()
```





<Figure size 1169x827 with 0 Axes>

```
plt.hist(df['salary'],edgecolor="black")
plt.title("Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Number of Employees")
plt.grid(axis='y', alpha=0.3)
plt.figure(figsize=(11.69,8.27))
plt.show()
```


Salary Distribution
Number of Employees
6
5
4
3
2
1